

PROFILE

A dedicated Geomatics Engineer Student and Satellite Data Analyst with a passion for GIS Mapping and Remote Sensing. Currently pursuing a Bachelor's degree in Geomatics Engineering at Kathmandu University. While working at Antarikchya Pratisthan Nepal I was experienced in satellite data processing, spatial analysis and map creation.

EDUCATION

Kathmandu University (KU) | Dhulikhel, Kavrepalanchok, Nepal *2025 – Expected 2029*

Bachelors of Engineering in Geomatics Engineering (Be-GE)

Website: geom.ku.edu.np/

Email: kushal7125@student.ku.edu.np

Key Modules: C Programming

Nepal Banepa Polytechnic Institute (NBPI) | Banepa-05, Kavrepalanchok, Nepal *2019 – 2023*

Diploma in Geomatics Engineering (DGE)

Website: nbpi.edu.np/

Aggregated Score: 77.86%

Key Modules: Surveying, Remote Sensing, Geographic Information Systems, Photogrammetry

Arunodaya English Secondary School | Panauti-4, Kavrepalanchok, Nepal *2006 – 2019*

Secondary Education Examination (SEE)

Aggregate Grade: 3.85

EXPERIENCE

Antarikchya Pratisthan Nepal | Lazimpath, Kathmandu, Nepal *August 2023 – September 2025*

Satellite Data Analyst | GIS Mapping

Website: antarikchya.org.np/

Email: kushal@antarikchya.org.np

Collaborative projects with Kathmandu Metropolitan City (KMC)

Asian Institute of technology | Paholyothin Highway, Klong Luang, Pathum Thani 12120

Thailand (remote)

March 2024 – August 2024

Remote Sensing Analyst | GIS Mapping

Collaborative projects with Dr. Ranadheer Mandadi

NAXA | Maharajgunj-3, Kathmandu, Nepal

June 2023 – July 2023

GIS Analyst

Website: naxa.com.np/

Email: kushal.naxa@gmail.com

Collaborative projects with Changunarayan Municipality and National Disaster Risk Reduction and Management Authority (NDRRMA)

PROJECTS & RESEARCH

Satellite Data Analytics Project (SDAP)

August 2023 – September 2025

Satellite Data Analyst | GIS Mapping

As part of the Satellite Data Analytics Project (SDAP), I have been actively engaged in leveraging satellite data and GIS mapping techniques for various critical tasks. My responsibilities include assessing flood damage in the Sapta Koshi, monitoring environmental changes in the Chure region, visualizing data related to earthquakes and forest fires, and conducting dynamic population density mapping. I have also contributed to the development of disaster protocols at Antarakchya Pratisthan, focusing on susceptibility mapping for floods and urban fires, utilizing Web GIS tools such as Leaflet.js, Mapbox, and MapTiler for map visualization purposes, while ensuring preparedness and effective response strategies.

Link: <https://antarikchya.org.np/visualization>

- ArcGIS Pro, QGIS, HEC-HMS, Google Earth Pro, Google Earth Engine (GEE), WebGIS, Php

Space-enabled Urban Solutions in Nepal (SUSN)

August 2023 – July 2024

Satellite Data Analyst | GIS Analyst

Playing a crucial role in the Space-enabled Urban Solutions in Nepal (SUSN) project, I've closely collaborated with Kathmandu Metropolitan City (KMC) departments, including the City Planning Commission (CPC), Disaster Management Department (DMD), Traffic Management and Road Safety Department, and Environment Management Department (EMD). Utilizing geospatial technologies, I've analyzed satellite data and conducted tailored land use classification. Working with both a dedicated team and various KMC departments, I've gained a comprehensive understanding of departmental needs. This understanding has enabled me to deliver GIS maps that facilitate informed decision-making for urban development and disaster risk management.

Link: <https://www.kmc.antarikchya.org.np/>

- ArcGIS Pro, QGIS, AutoCad, Google Earth Pro, Google Earth Engine (GEE), Python

Land Use/Land Cover Classification and Future Prediction

March 2024 – August 2024

Remote Sensing Analyst | GIS Mapping

Engaged in collaborative projects with Ranadheer Reddy at the Asian Institute of Technology in Thailand. Specifically involved in the Land Use/Land Cover Classification and Future Prediction project, utilizing Google Earth Engine (GEE) for LULC mapping and TerrSet 2020 for predicting future LULC changes over time.

- ArcGIS Pro, Google Earth Engine (GEE), TerrSet2020

Changunarayan Municipality

June 2023 – July 2023

GIS Analyst

Contributed to the digitization project in Changunarayan Municipality, assisting in the manual digitization of buildings using high-resolution drone data. Additionally, participated in the road network analysis to determine road directions, ensuring accurate spatial data for urban planning.

Link: <https://changu-stag.naxa.com.np/>

- ArcGIS Pro, QGIS

National Building Exposure

June 2023 – July 2023

Contributed to the National Building Exposure project by working as a building digitizer. Implemented manual digitization techniques to accurately capture building footprints, contributing to the creation of a comprehensive building exposure database.

Link : <https://exposure.ndrrma.gov.np/>

- ArcGIS

Impact on COR Station due to Gorkha Earthquake 2015

November 2022 – March 2023

GIS Analyst

This is the final year project for my Diploma degree, I utilized UNAVCO to gather control point data, which was then processed. The processed data was visualized using ArcGIS. The project focused on investigating the impact of the Gorkha Earthquake in 2015 on COR Station, specifically identifying the shift of control points caused by the Gorkha Earthquake 2015.

- ArcGIS, UNAVCO

Topographical and Cadastral Surveying | Jiri, Nepal

November 23 – December 02, 2023

Participated in topographical and cadastral surveying in Jiri, Nepal. Utilized DGPS to establish control points and employed total stations and prisms for various surveying tasks, including projects related to hydropower, routes, transmission lines, bridges, and canals. Processed the collected data and visualized it using ArcGIS.

- ArcGIS, Total Station (TS), Differential Global Positioning Systems (DGPSs), Prism, Measuring Tape

CONFERENCE AND PAPERS

"Monsoon Flood Impact Analysis of Nepal's most populous and Capital City, Kathmandu"
and "Sapta Koshi Flood: Turning Risk into a valuable opportunity"

Innovation and Transformation in the Development Effort through Science, Technology and Governance | Kathmandu, Nepal

May 17 – 19, 2024

Link: <https://www.mitrakunj.org.np/international-conference/>

NYC Scientists' Conference 2081 | Jawalakhel, Lalitpur

June 19, 2024

Link: <https://www.ran.org.np/>

PAPERS AND PUBLICATIONS

Leveraging Remote Sensing for Lineament Extraction from Landsat-8, Sentinel-2 and Sentinel-1 at Bagmati Province of Nepal *25 November 2025*

Link: <https://ieeexplore.ieee.org/document/11242632>

The Cubesat Case Study: Resource- Efficient Deep Learning for Space Application

Link: <https://ieeexplore.ieee.org/document/11314016>

25 November 2025

TRAINING

Google Earth Engine - Hands-on skills in satellite data analysis and geospatial processing using the cloud-based Earth Engine platform.

Basic QGIS - How to visualize, analyze, and create maps using QGIS, including adding layers, styling maps, and performing basic spatial analysis like clip, buffer.

HEC-RAS & HEC-HMS - Performed rainfall-runoff modeling using HEC-HMS and simulated flood inundation maps using HEC-RAS for hydrologic and hydraulic analysis.

CERTIFICATIONS AND PARTICIPATIONS

- Python Foundation for Spatial Analysis *December 02 – 10, 2025*
- Google Earth Engine for Water Resources Management *August 04 – 20, 2025*
- Space Con 2025, Kathmandu, Nepal *April 11, 2025*
- Secured the 1st position in the Map Design Competition:
<https://www.facebook.com/gesan2014/posts/pfbid0caDmbQWuUj4EXBAw53xPiBRvfcak6912kULtaV587jv3hWRTGavZjE9MXTQL88srl> *June 27, 2024*
- Honorable Mention for Compelling Visuals winner for NASA's Pale Blue Dot: Visualization Challenge:
<https://drivendata-public-assets.s3.amazonaws.com/nasa-open-science-gallery/nayan.bakhadyo/visual.png> *March 14, 2024*
- Training on Google Earth Engine *January 18 – May 02, 2024*
- Training on PyQGIS Masterclass *July 15 – August 9, 2024*
- Training on Advanced 30DayOfQGIS *May 13 – June 11, 2024*
- Space Con 2024, Lalitpur, Nepal *April 12, 2024*
- Training on Land Use & GIS for Technical Personnel of Municipalities within Kathmandu Valley *January 29, 2024 – February 2, 2024*
- Training on Web GIS *December 2023 – February 2024*
- Esri MOOC *2023*
- Training on AutoCad *November 07 – 08, 2023*
- Training on Python *June 16 – 26, 2023*
- Training on OpenStreetMap *August 17 – 18, 2022*

SKILLS

- Project management
- ArcGIS Pro
- WebGIS
- Autocad
- HEC-HMS
- ERDAS Imagine
- SNAP Desktop
- Microsoft Office
- TerrSet 2020
- Total Station
- Differential Global Positioning Systems
- Communication
- QGIS
- ArcGIS
- Google Earth Engine
- HEC-RAS
- ENVI
- Python
- HTML/CSS/JS
- PHP
- Theodolite
- Plane Table

WORK REFERENCES

- **Dr. Abhas Maskey**
 - Founder | Managing Director, Antarikchya Pratisthan Nepal
 - Email: abhas@antarikchya.org.np
 - Phone: +977 9869397707
- **Prof. Dr.-Ing. Ramesh Kumar Maskey**
 - Academician, Nepal Academy of Science and Technology (NAST)
 - Email: rmaskey@ku.edu.np
 - Phone: +977 9851102669
- **Er. Anukul Nidhi**
 - Assistant Traffic Engineer, City Planning Commission, Kathmandu Metropolitan City
 - Email: nidhianukul479@gmail.com
 - Phone: +977 9844417900
- **Er. Sujan Nepali**
 - Geospatial Consultant at ICIMOD | Geomatics Instructor at NBPI
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